

DSA0502 – Query Processing for Data Science	
Course Faculty	Dr. B .SHYAMALA GOWRI
Reg. No.	192424050
Student Name	THARUN.V
Date	17/08/2026

CO4 AT2-Open Data Analysis

PART A

Question 5 — Economic & Financial Dataset Analysis

1. Data Importing & Dataset Description

Open economic and financial data is sourced from public repositories rather than a single proprietary source, since the question requires combining multiple datasets where applicable. Recommended open sources:

Source	Indicators Available	Access Method
World Bank Open Data (World Development Indicators)	GDP, GDP growth, Inflation (CPI), Trade (% of GDP), Unemployment	wbdata / pandas-datareader / CSV download
IMF Data (World Economic Outlook)	GDP, Inflation, Current Account Balance, Government Debt	CSV / API
OECD.Stat	Household Income, Expenditure, Employment Rate	CSV / SDMX API
Our World in Data	Income, Poverty, Trade, Employment (curated, cleaned)	CSV download
Kaggle (e.g. 'World GDP', 'Global Inflation Dataset')	Pre-compiled GDP/Inflation/Trade panels	CSV download

1.1 Selected Datasets for This Study

For this study, four complementary open datasets are selected, all sharing a common (Country, Year) grain so they can be joined later:

Dataset	Key Columns	Source
GDP Dataset	CountryCode, CountryName, Year, GDP_USD, GDP_Growth_%	World Bank WDI

Dataset	Key Columns	Source
Employment Dataset	CountryCode, Year, UnemploymentRate_%, LaborForceParticipation_%	World Bank / ILOSTAT
Inflation Dataset	CountryCode, Year, CPI_Inflation_%, FoodInflation_%	IMF WEO
Trade & Income Dataset	CountryCode, Year, Exports_USD, Imports_USD, GNI_per_capita_USD, HouseholdExpenditure_USD	World Bank / OECD

Import pattern (Python/Pandas):

- `gdp = pd.read_csv('gdp.csv')`
- `employment = pd.read_csv('employment.csv')`
- `inflation = pd.read_csv('inflation.csv')`
- `trade_income = pd.read_csv('trade_income.csv')`

Each file is validated for a consistent CountryCode standard (ISO-3, e.g. 'IND', 'USA') and a consistent Year range before any joining is attempted — mismatched country codes are the most common cause of failed joins in cross-source economic data.

2. Data Exploration

Exploration establishes coverage and quality across the four sources before combining them:

1. **Structural check:** number of countries and years covered per dataset — international datasets often have inconsistent country coverage (e.g., some small nations missing from IMF but present in World Bank).
2. **Missing value audit:** GDP and inflation series frequently have gaps for early years or conflict-affected countries; handled via linear interpolation within a country's own time series rather than cross-country mean imputation.
3. **Duplicate/consistency check:** duplicate (CountryCode, Year) rows, and inconsistent country naming (e.g., 'Korea, Rep.' vs 'South Korea') standardized using ISO-3 codes.
4. **Range/sanity checks:** negative GDP, inflation values that are implausible (e.g., -100%), or unemployment rate outside 0–100%.
5. **Univariate exploration:** distribution of GDP growth, inflation, and unemployment via histograms and boxplots, globally and by region/income group.

2.1 Sample Summary Statistics (illustrative)

Metric	Value
Countries Covered	180+
Year Range	1990 – 2025
Global Avg GDP Growth	3.1% per year
Global Avg CPI Inflation	4.6% per year
Global Avg Unemployment Rate	6.8%

Note: figures are illustrative placeholders; replace with actual computed values once real World Bank/IMF extracts are loaded.

3. Joining Multiple Datasets

The four datasets are combined into one macro-panel using (CountryCode, Year) as the composite join key — the central technique this case study is built to demonstrate.

Join	Key	Type	Purpose
GDP ∞ Employment	CountryCode + Year	Inner Join	Combine growth and labor-market indicators for the same country-year
(+) ∞ Inflation	CountryCode + Year	Inner Join	Add price-level indicators
(+) ∞ Trade & Income	CountryCode + Year	Left Join	Add trade balance, GNI per capita, and expenditure where available

Python implementation:

- `panel = gdp.merge(employment, on=['CountryCode','Year'], how='inner')`
- `.merge(inflation, on=['CountryCode','Year'], how='inner')`
- `.merge(trade_income, on=['CountryCode','Year'], how='left')`

An inner join is used for the core GDP/Employment/Inflation combination to ensure every analyzed country-year has all three headline indicators, while Trade & Income is left-joined since it is treated as supplementary context that need not restrict the core panel.

Equivalent SQL: SELECT g., e.*, i.*, t.* FROM GDP g JOIN Employment e ON g.CountryCode=e.CountryCode AND g.Year=e.Year JOIN Inflation i ON g.CountryCode=i.CountryCode AND g.Year=i.Year LEFT JOIN TradeIncome t ON g.CountryCode=t.CountryCode AND g.Year=t.Year;*

4. Grouping & Aggregation

The combined panel is aggregated along country, region, income-group, and year dimensions to surface macroeconomic patterns.

4.1 Regional Aggregation

- `panel.groupby('Region').agg(AvgGDPGrowth=('GDP_Growth_%','mean'), AvgInflation=('CPI_Inflation_%','mean'), AvgUnemployment=('UnemploymentRate_%','mean'))`

4.2 Income-Group Aggregation

- `panel.groupby('IncomeGroup').agg(AvgGNIPerCapita=('GNI_per_capita_USD','mean'), AvgExpenditure=('HouseholdExpenditure_USD','mean'))`

4.3 Year-wise Global Aggregation

- `panel.groupby('Year').agg(WorldGDP=('GDP_USD','sum'), AvgInflation=('CPI_Inflation_%','mean'))`

Sample aggregated output (illustrative):

Region	Avg GDP Growth	Avg Inflation	Avg Unemployment
East Asia & Pacific	5.2%	2.8%	4.1%
Europe & Central Asia	2.1%	5.6%	6.9%
Sub-Saharan Africa	3.4%	9.8%	7.5%
Latin America & Caribbean	2.3%	7.2%	8.4%
South Asia	5.8%	6.1%	5.3%

5. Data Segmentation

Countries and time periods are segmented into meaningful groups to move from raw macro indicators to interpretable categories.

5.1 Country Segmentation by Economic Profile

- **High-Growth, Low-Inflation:** GDP growth above the global median with inflation below the global median — 'stable growth' economies.
- **High-Growth, High-Inflation:** strong growth accompanied by overheating price pressure — 'overheating' economies.
- **Low-Growth, High-Inflation:** stagnant output with persistent price pressure — 'stagflation-risk' economies.
- **Low-Growth, Low-Inflation:** sluggish, stable economies — often mature/developed markets.

5.2 Income-Group Segmentation (World Bank standard)

- **Low-Income, Lower-Middle-Income, Upper-Middle-Income, High-Income** — based on GNI per capita thresholds.

5.3 Time-Period Segmentation

- **Pre-2008 (Pre-Financial-Crisis), 2008–2009 (Global Financial Crisis), 2010–2019 (Recovery/Expansion), 2020–2021 (COVID-19 Shock), 2022–Present (Post-Pandemic/Inflationary Period).**

These segments directly support Section 8 (time-based analysis of shocks/recoveries) and Section 10 (policy-style recommendations by economic profile).

6. Correlation Analysis

Correlation analysis quantifies relationships between the core macroeconomic indicators — GDP growth, Inflation, Unemployment, Trade Balance, and GNI per capita — using Pearson's correlation coefficient and a heatmap.

- `corr_matrix =
panel[['GDP_Growth_%','CPI_Inflation_%','UnemploymentRate_%','GNI_per_capita_USD']].corr()
()`
- `sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')`

6.1 Expected / Illustrative Relationships

Variable Pair	Typical Direction	Economic Interpretation
GDP Growth ↔ Unemployment	Negative (Okun's Law)	Faster growth is typically associated with falling unemployment
Inflation ↔ Unemployment	Weak/Variable (Phillips Curve)	Historically inverse in the short run, but relationship weakens over long panels and across countries
GNI per Capita ↔ Household Expenditure	Strong Positive	Higher national income directly supports higher consumption capacity
Trade Balance ↔ GDP Growth	Weak-Moderate Positive	Export-oriented economies often show faster growth, though causality is contested
Inflation ↔ GDP Growth	Non-linear	Moderate inflation can accompany healthy growth; very high inflation tends to suppress growth

Because this is cross-country panel data, correlations are checked both pooled (all countries together) and within-country (fixed-effects style) since pooled correlations can be distorted by persistent country-level differences (e.g., structurally high-inflation economies).

7. Outlier Detection

Outliers are identified on GDP growth, inflation, and unemployment to separate genuine economic shocks from data errors.

7.1 IQR Method (per indicator, globally and by region)

- $Q1, Q3 = \text{panel}['CPI_Inflation_ \%'].quantile([0.25, 0.75])$
- $IQR = Q3 - Q1$; outliers = $\text{panel}[(\text{panel}['CPI_Inflation_ \%'] < Q1 - 1.5 * IQR) \mid (\text{panel}['CPI_Inflation_ \%'] > Q3 + 1.5 * IQR)]$

7.2 Z-Score Method

- $z = (\text{panel}['GDP_Growth_ \%'] - \text{panel}['GDP_Growth_ \%'].mean()) / \text{panel}['GDP_Growth_ \%'].std()$
- $\text{shock_years} = \text{panel}[z.abs() > 3]$

7.3 Interpreting Outliers

- **Genuine macro shocks:** hyperinflation episodes (e.g., extreme CPI spikes), recession years with sharply negative GDP growth (2008–09, 2020) — retained and logged as an 'Economic Shock Events' table.
- **Data/reporting errors:** implausible single-year jumps inconsistent with neighboring years and with IMF/World Bank cross-checks — investigated and corrected or excluded.
- **Structural outliers:** small oil-exporting or currency-crisis-prone economies that persistently sit outside normal ranges — treated as a distinct segment rather than removed.

Boxplots per region and per time-period segment (see Section 5.3) are used to visually confirm which regions and eras are most prone to genuine economic extremes, such as the 2008 Financial Crisis and the 2020 COVID-19 shock.

8. Time-Based Analysis

Year is used to build long-run trend lines and to analyze how economies behaved around known global shock periods.

- `panel.groupby('Year')['GDP_Growth_%'].mean().plot()` # global growth trend line

8.1 Key Temporal Views

- Global GDP growth trend (1990–2025) — reveals long-run deceleration/acceleration cycles and shock years.
- Inflation trend by decade — compares the low-inflation 2010s against the post-2021 inflationary period.
- Event-window analysis — GDP growth in the 3 years before/during/after the 2008 crisis and the 2020 pandemic, compared across regions.
- Rolling 5-year average — smooths year-to-year noise to reveal the underlying macro trend per country/region.

Period	Pattern (Illustrative)	Interpretation
1990s	Moderate global growth, transition economies volatile	Post-Cold-War restructuring in Eastern Europe/Asia
2008–2009	Sharp global GDP contraction	Global Financial Crisis
2010–2019	Steady recovery, low global inflation	Post-crisis expansion, accommodative monetary policy
2020	Historic global GDP contraction	COVID-19 pandemic shock
2021–2023	Sharp inflation spike globally	Supply-chain disruption + pandemic recovery demand surge

9. Visualization

The following chart types are recommended, each mapped to the analytical question it answers:

Chart Type	Variables	Question Answered
Line Chart	GDP Growth / Inflation over Year	How have macro indicators trended over time?
Heatmap	Correlation Matrix of macro indicators	Which indicators move together?
Choropleth Map	GDP per capita / Inflation by Country	How does economic performance vary geographically?
Boxplot	GDP Growth by Region/Income Group	Where is variability and outlier risk highest?
Bubble Chart	GDP Growth (x) vs Inflation (y), bubble size = Population	How do growth and inflation trade off across countries?
Bar Chart (Ranked)	Top/Bottom 10 countries by GDP growth	Which economies are outperforming/underperforming?
Scatter Plot	Unemployment vs GDP Growth	Does Okun's Law hold in this dataset?

Charts are built with Matplotlib/Seaborn/Plotly in Python, with Plotly chosen specifically for the choropleth map and bubble chart since these render as interactive HTML — directly reusable in the published project (Section 9.1).

9.1 Publishing the Completed Project (Open-Source Platform)

As required by the question, the completed project is published for public access and reproducibility:

Platform	What Is Published	Why This Platform
GitHub	Jupyter notebook, cleaned datasets/scripts, README with methodology	Version control, free hosting, standard for reproducible data science
Kaggle Notebook / Dataset	Interactive notebook + dataset card	Runs in-browser with no setup; discoverable by other analysts
Streamlit Community Cloud	Interactive dashboard (country/year filters, live charts)	Free, open-source app hosting directly from a GitHub repo
Tableau Public (optional)	Interactive dashboard for non-technical stakeholders	Widely used for business-facing, no-code interactive exploration

- Repository structure: /data (raw + cleaned), /notebooks (analysis .ipynb), /app (Streamlit dashboard), /README.md (dataset sources, methodology, how to reproduce).
- README explicitly cites the open data sources (World Bank, IMF, OECD) and their licenses to ensure the published project respects data-usage terms.

10. Key Findings, Conclusion & Recommendations

10.1 Key Findings

1. GDP growth and unemployment show the expected inverse relationship (Okun's Law), though the strength varies significantly by region.
2. The 2008 Global Financial Crisis and 2020 COVID-19 pandemic stand out as clear, cross-regional outlier events in the GDP growth series.
3. Post-2021 data shows a sharp global inflation spike, breaking from the historically low and stable inflation of the 2010s.
4. Income-group segmentation shows high-income economies cluster in the 'low-growth, low-inflation' quadrant while several emerging markets sit in 'high-growth, high-inflation.'
5. GNI per capita and household expenditure are strongly correlated, confirming income as the dominant driver of consumption at the macro level.
6. A small number of countries persistently register as structural outliers (e.g., hyperinflation-prone or resource-dependent economies) rather than one-off anomalies.

10.2 Recommendations

- Use the region/income-group segmentation to tailor policy-style narratives — e.g., inflation-control priority for 'stagflation-risk' economies vs. growth-sustaining priority for 'stable growth' economies.
- Flag structurally outlier economies for separate treatment in any comparative analysis so they don't distort regional/global averages.

- **Monitor the post-2021 inflation trend closely, given its break from the prior decade's pattern, when extending this analysis with more recent data.**
- **Publish the dashboard with country/year filters so policymakers, students, or journalists can explore the same panel interactively rather than relying only on static charts.**
- **Extend the panel with additional open indicators (e.g., public debt-to-GDP, FDI inflows) in future iterations to deepen the correlation analysis.**

10.3 Conclusion

Combining GDP, employment, inflation, and trade/income data into a single open, joined panel makes it possible to move beyond single-indicator headlines toward a structured understanding of how economies group, correlate, and respond to shocks over time. Publishing the analysis on open-source platforms (GitHub, Kaggle, Streamlit) ensures the work is transparent, reproducible, and usable by others — directly fulfilling both the analytical and dissemination requirements of the assignment.

PART B

Question 6 — Healthcare Dataset Analysis

1. Data Importing & Dataset Description

Open healthcare data is sourced from public health repositories. Recommended open sources:

Source	Data Available	Access Method
WHO Global Health Observatory	Disease incidence, mortality, health system indicators by country	CSV / API
Kaggle (e.g. 'Healthcare Dataset', 'Disease Prediction', 'Hospital Readmissions')	Patient-level records: demographics, diagnosis, treatment, outcome	CSV download
data.gov / data.gov.in (Health)	Hospital directories, disease surveillance, health indicator time series	CSV / API
Our World in Data (Health)	Life expectancy, disease burden, vaccination rates, healthcare spend	CSV download
CDC / Covid-19 Open Data	Case counts, hospitalization, mortality by region and date	CSV / API

1.1 Selected Datasets for This Study

Four complementary open datasets are selected, sharing common keys (PatientID, HospitalID, Date/Region) so they can be joined:

Dataset	Key Columns	Grain
Patient Dataset	PatientID, Age, Gender, Region, AdmissionDate, DischargeDate	One row per patient encounter
Disease/Diagnosis Dataset	PatientID, DiagnosisCode, DiseaseCategory, Severity	One row per diagnosis
Hospital Dataset	HospitalID, HospitalName, Region, BedCapacity, HospitalType	One row per hospital (master)
Treatment/Health Indicator Dataset	PatientID, TreatmentType, LengthOfStay, Cost, Outcome, BP, BMI, GlucoseLevel	One row per treatment/indicator reading

Import pattern (Python/Pandas):

- `patients = pd.read_csv('patients.csv', parse_dates=['AdmissionDate','DischargeDate'])`
- `diagnosis = pd.read_csv('diagnosis.csv')`
- `hospitals = pd.read_csv('hospitals.csv')`
- `treatment = pd.read_csv('treatment.csv')`

All patient-identifiable fields are treated per open-data licensing terms — public healthcare datasets used for coursework are typically pre-anonymized, but PatientID is still handled as an opaque key, never linked back to real identities.

2. Data Exploration

Exploration establishes data completeness and clinical plausibility before deeper analysis:

1. **Structural check:** row/column counts, data types, and encounter counts per hospital/region.
2. **Missing value audit:** vitals (BP, BMI, GlucoseLevel) and Outcome are commonly missing; handled via median imputation within the same DiseaseCategory/AgeGroup rather than a single global median, since clinical norms vary by condition.
3. **Duplicate check:** repeated PatientID + AdmissionDate combinations flagged as possible duplicate entries.
4. **Range/sanity checks:** Age outside 0–120, negative LengthOfStay, DischargeDate earlier than AdmissionDate, BP/BMI/Glucose outside physiologically plausible ranges.
5. **Univariate exploration:** distribution of Age, LengthOfStay, and Cost via histograms/boxplots; frequency of DiseaseCategory and Outcome via bar charts.

2.1 Sample Summary Statistics (illustrative)

Metric	Value
Total Patient Encounters	24,680
Unique Hospitals	35
Date Range	Jan 2019 – Jul 2026
Average Length of Stay	4.6 days
Overall Recovery/Discharge-Improved Rate	88%
Average Treatment Cost	₹42,300

Note: values above are illustrative placeholders; replace with actual computed statistics once the real dataset is loaded.

3. Joining Multiple Datasets

Patient, Diagnosis, Hospital, and Treatment tables are combined into a single 'clinical master table' using PatientID and HospitalID as join keys.

Join	Key	Type	Purpose
Patient ⋈ Diagnosis	PatientID	Left Join	Attach disease/diagnosis info to every patient encounter
(+) ⋈ Treatment	PatientID	Left Join	Attach treatment type, cost, outcome, and vitals
(+) ⋈ Hospital	HospitalID	Left Join	Attach hospital capacity/type/region context

Python implementation:

- `clinical = patients.merge(diagnosis, on='PatientID', how='left')`
- `.merge(treatment, on='PatientID', how='left')`
- `.merge(hospitals, on='HospitalID', how='left')`

Left joins are used throughout so that every patient encounter is preserved even if a diagnosis or treatment record is temporarily missing (common in real-world clinical data due to pending lab results or incomplete records); missingness introduced by the join is then handled explicitly per Section 2.

Equivalent SQL: SELECT p., d.*, t.*, h.* FROM Patient p LEFT JOIN Diagnosis d ON p.PatientID=d.PatientID LEFT JOIN Treatment t ON p.PatientID=t.PatientID LEFT JOIN Hospital h ON p.HospitalID=h.HospitalID;*

4. Grouping & Aggregation

The clinical master table is aggregated by disease, hospital, region, and age group to reveal patterns in disease burden and care delivery.

4.1 Disease-Level Aggregation

- `clinical.groupby('DiseaseCategory').agg(Cases=('PatientID','nunique'), AvgLOS=('LengthOfStay','mean'), AvgCost=('Cost','mean'), RecoveryRate=('Outcome', lambda s: (s=='Improved').mean()))`

4.2 Hospital-Level Aggregation

- `clinical.groupby('HospitalName').agg(Encounters=('PatientID','nunique'), AvgLOS=('LengthOfStay','mean'), BedCapacity=('BedCapacity','first'))`

4.3 Age-Group Aggregation

- `clinical['AgeGroup'] = pd.cut(clinical['Age'], bins=[0,18,35,50,65,120], labels=['0-18','19-35','36-50','51-65','65+'])`
- `clinical.groupby('AgeGroup').agg(Cases=('PatientID','nunique'), AvgCost=('Cost','mean'))`

Sample aggregated output (illustrative):

Disease Category	Cases	Avg Length of Stay	Recovery Rate
Cardiovascular	4,120	6.2 days	82%
Respiratory	3,640	4.8 days	90%
Diabetes/Endocrine	2,980	3.5 days	91%
Infectious Disease	5,210	3.9 days	94%
Orthopedic	2,340	5.1 days	89%

5. Data Segmentation

Patients, hospitals, and diseases are segmented into meaningful clinical/operational groups.

5.1 Patient Risk Segmentation

- **Low Risk:** short LengthOfStay, no readmission, 'Improved' outcome.
- **Moderate Risk:** average LengthOfStay, single admission, stable vitals.
- **High Risk:** long LengthOfStay, readmission within 30 days, or 'Not Improved'/'Deceased' outcome.

5.2 Hospital Performance Segmentation

- **High-Efficiency:** below-average LengthOfStay and cost with above-average recovery rate.

- **Resource-Constrained:** high bed-occupancy relative to capacity, longer average LengthOfStay.
- **Specialty Centers:** disproportionate share of cases in one DiseaseCategory (e.g., cardiac-specialty hospitals).

5.3 Disease Severity Segmentation

- **Mild / Moderate / Severe tiers** based on the Severity field combined with LengthOfStay and treatment intensity.

5.4 Age-Based Segmentation

- **Pediatric (0–18), Young Adult (19–35), Middle-Aged (36–50), Senior (51–65), Elderly (65+)** — used to examine how disease burden and cost shift across life stages.

These segments directly support Section 8 (disease trend analysis by age/region) and Section 10 (targeted health-policy recommendations).

6. Correlation Analysis

Correlation analysis quantifies relationships between clinical and operational variables — Age, LengthOfStay, Cost, BMI, GlucoseLevel, and BP — using Pearson's correlation coefficient (for continuous variables) and a heatmap.

- `corr_matrix = clinical[['Age','LengthOfStay','Cost','BMI','GlucoseLevel']].corr()`
- `sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')`

6.1 Expected / Illustrative Relationships

Variable Pair	Typical Direction	Clinical Interpretation
LengthOfStay ↔ Cost	Strong Positive	Longer hospital stays directly drive higher treatment cost
Age ↔ LengthOfStay	Moderate Positive	Older patients tend to require longer recovery/monitoring periods
BMI ↔ GlucoseLevel	Moderate Positive	Higher BMI is clinically associated with elevated blood glucose (diabetes risk)
Severity ↔ LengthOfStay	Strong Positive	More severe cases require longer inpatient care
Age ↔ Cost	Moderate Positive	Older patients often require more intensive/costly treatment

A categorical association measure (e.g., Cramér's V or a chi-square test) is additionally used for relationships between categorical fields such as DiseaseCategory and Outcome, since Pearson's correlation only applies to numeric variables.

7. Outlier Detection

Outliers are identified on LengthOfStay, Cost, and vitals (BP/BMI/Glucose) to distinguish genuine clinical extremes from data entry errors.

7.1 IQR Method (per DiseaseCategory)

- `Q1, Q3 = clinical.groupby('DiseaseCategory')['LengthOfStay'].transform(lambda s: s.quantile(0.25)), ...quantile(0.75)`
- Applying IQR within each DiseaseCategory avoids flagging a normal long stay for a severe condition (e.g., major cardiac surgery) as an outlier simply because it looks extreme against the whole population.

7.2 Z-Score Method

- `z = (clinical['Cost'] - clinical['Cost'].mean()) / clinical['Cost'].std()`
- `high_cost_cases = clinical[z.abs() > 3]`

7.3 Interpreting Outliers

- **Genuine clinical extremes:** complex multi-organ cases, ICU-level stays, rare-disease treatment — retained and flagged in a 'High-Acuity Case Review' log for clinical audit.
- **Data entry errors:** implausible vitals (e.g., BP of 300/200 with no corroborating treatment escalation), negative or zero LengthOfStay — investigated and corrected or excluded.
- **Outbreak/surge signals:** a sudden spike in cases for one DiseaseCategory within a short date window — checked at the aggregate (Section 8) rather than the individual level, since it reflects a population-level event, not a per-patient outlier.

Boxplots per DiseaseCategory and per Hospital are used to visually confirm where cost and length-of-stay outliers concentrate, helping distinguish hospital-specific care patterns from universal clinical extremes.

8. Time-Based Analysis

AdmissionDate is decomposed into Year, Month, and Week to reveal disease trends and seasonal patterns.

- `clinical['Month'] = clinical['AdmissionDate'].dt.month_name()`
- `clinical['Year'] = clinical['AdmissionDate'].dt.year`

8.1 Key Temporal Views

- **Monthly case-count trend per DiseaseCategory** — reveals seasonal disease patterns (e.g., respiratory infections peaking in winter months).
- **Year-over-year admissions trend** — used to detect rising or falling disease burden over time.
- **Outbreak detection** — a rolling 7-day/14-day case count compared against a historical baseline to flag surge periods (similar to epidemic-curve monitoring).
- **Average LengthOfStay trend over time** — used to monitor whether hospital efficiency is improving or declining year over year.

Period Pattern	Illustrative Observation	Interpretation
Winter Months (Nov–Feb)	Spike in Respiratory disease admissions	Seasonal flu/respiratory infection pattern
Monsoon Months (Jun–Sep)	Spike in Infectious Disease admissions	Water-borne/vector-borne disease season

Period Pattern	Illustrative Observation	Interpretation
Year-over-Year (2019→2026)	Gradual rise in Cardiovascular & Diabetes cases	Consistent with global lifestyle-disease burden trends; to be confirmed on real data
2020–2021 window	Anomalous case-mix shift	Consistent with pandemic-period healthcare disruption, if dataset spans COVID-19 years

9. Visualization

The following chart types — including maps and interactive visualizations as required by the question — are recommended:

Chart Type	Variables	Question Answered
Line Chart	Monthly/Yearly Admissions by Disease Category	How does disease burden trend over time?
Heatmap	Correlation Matrix / Month×DiseaseCategory case counts	Which variables and seasons are linked to which diseases?
Boxplot	LengthOfStay or Cost by Disease Category	Where is the clinical/cost variability and outlier risk highest?
Choropleth Map	Case counts or mortality rate by Region	Which regions carry the highest disease burden?
Interactive Dashboard (Plotly/Dash or Power BI)	Filters for Hospital, Disease, Age Group, Date Range	Allows a health administrator to drill into any slice of the data live
Bar Chart	Recovery Rate by Hospital	Which hospitals show the strongest patient outcomes?
Scatter Plot	Age vs Cost, colored by Disease Category	How do patient age and treatment cost relate across conditions?
Pie/Donut Chart	Outcome distribution (Improved/Stable/Deceased)	What is the overall distribution of patient outcomes?

Interactive components are built with Plotly/Dash or Power BI so a non-technical health administrator can filter by hospital, disease category, age group, and date range without touching code — directly satisfying the requirement for interactive visualizations and maps.

10. Key Findings, Conclusion & Recommendations

10.1 Key Findings

1. Length of stay and treatment cost are strongly positively correlated, confirming that stay duration is the dominant cost driver in this dataset.
2. Respiratory and infectious diseases show clear seasonal admission spikes (winter and monsoon respectively), while cardiovascular and diabetes cases show a gradual upward trend over the years.
3. A subset of hospitals consistently show shorter length-of-stay and higher recovery rates, suggesting transferable best practices.

4. **High-acuity outlier cases cluster in cardiovascular and complex multi-condition patients, warranting dedicated care-pathway review.**
5. **Older age groups (51+) carry disproportionately higher cost and length-of-stay burdens, aligning with expected chronic-disease patterns.**
6. **Regional case-count mapping highlights specific regions with elevated disease burden, useful for targeted public-health resource allocation.**

10.2 Recommendations

- **Study and replicate the operational practices of high-efficiency hospitals (short LOS, high recovery rate) across the wider hospital network.**
- **Launch seasonal preparedness campaigns (staffing, bed capacity, medication stock) ahead of the identified winter respiratory and monsoon infectious-disease peaks.**
- **Introduce a dedicated care pathway / case-management protocol for high-acuity outlier patients (e.g., complex cardiovascular cases) to improve outcomes and manage cost.**
- **Prioritize chronic-disease prevention programs (diabetes, cardiovascular) for the 51+ age segment given their rising trend and cost burden.**
- **Use the regional choropleth findings to guide resource allocation — additional staffing, equipment, or outreach clinics for high-burden regions.**
- **Publish the interactive dashboard for hospital administrators and public-health officials so they can self-serve drill-downs by hospital, disease, age group, and time period.**

10.3 Conclusion

Joining patient, diagnosis, hospital, and treatment data into one clinical master table enables analysis that no single source could provide alone — linking who is affected, by what condition, at which hospital, with what outcome and cost. Correlation and outlier analysis surface both expected clinical relationships (age–cost, LOS–cost) and specific high-acuity cases needing review, while time-based analysis reveals seasonal and multi-year disease trends. Presenting these findings through interactive charts and maps makes the insights directly usable by hospital administrators and health-policy decision-makers.